

QORLA Stability Test 3

End-to-End Scaling and Comparative Stability

Complete Experimental Report

Sequence length, depth, width, multi-head structure, precision, optimization stress, controlled attention comparison, and final blind multi-split confirmation

QORLA Research Series

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Status: STAB3 closed; SG10 frozen; no further tuning permitted within STAB3.

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Executive Summary

Stability Test 3 (STAB3) was designed to answer a different question from the earlier mechanistic studies: once the QORLA instability mechanism had been identified, could a fixed training solution survive systematic scaling and controlled comparison without being continually retuned? The phase began with the permanent correction stop-gradient intervention inherited from STAB2, discovered that permanent detachment imposed a capacity cost in otherwise stable regimes, and then derived a transient forward-exact correction-gradient schedule, SG10. That schedule was frozen after STAB3-02D and used without further schedule tuning in all later STAB3 experiments.

The central frozen QORLA forward operator remained

$$o_i = B_i + R_i, \quad (1)$$

while the training-time backward rule became

$$o_i^{\text{train}} = B_i + \text{sg}(R_i) + \alpha(f) [R_i - \text{sg}(R_i)], \quad (2)$$

with

$$\alpha(f) = \begin{cases} 0, & f \leq 0.10, \\ \frac{f - 0.10}{0.20}, & 0.10 < f \leq 0.30, \\ 1, & f > 0.30. \end{cases} \quad (3)$$

Thus the correction contributes fully to the forward pass at every point in training, but its gradient is suppressed for the first 10% of training, smoothly restored from 10–30%, and fully restored thereafter.

Across STAB3, this rule showed strong benefits in many, but not all, regimes. It transferred across depth, moderate width, multi-head architectures, BF16 and FP16 training, standard batch sizes, and targeted Q/K initialization stress. It also exhibited clear boundaries: aggressive learning rates, very small batch sizes, and some very-wide trajectories remained unstable. These failures were retained as part of the result rather than tuned away.

The final predeclared multi-split confirmation compared three exactly parameter-matched controlled models: DIST-SOFTMAX, QORLA-BASE, and QORLA-SG10. Across nine split/seed pairs, QORLA-SG10 reached $91.574 \pm 1.667\%$ mean final accuracy, compared with $89.969 \pm 1.709\%$ for DIST-SOFTMAX and $81.235 \pm 11.719\%$ for QORLA-BASE. QORLA-SG10 beat QORLA-BASE in 9/9 pairs with a mean gain of +10.340 percentage points, and beat DIST-SOFTMAX in 7/9 pairs with a mean gain of +1.605 percentage points. However, the controlled QORLA implementation was slower and used more memory than the distance-softmax control. Therefore STAB3 supports an optimization-and-representation claim, not a systems-efficiency claim.

Final STAB3 conclusion. The local QORLA correction pathway contains useful modeling capacity, but unrestricted early gradient propagation through that pathway can generate severe trajectory-dependent instability. A forward-exact transient correction-gradient schedule substantially suppresses that instability in a broad operating region while eventually restoring ordinary QORLA training. The benefit is real but bounded, and the present evidence remains controlled rather than real-world or LLM-scale. Those questions are reserved for STAB4.

1 Purpose and Experimental Philosophy

STAB2 established that deep QORLA failures were causally linked to backward propagation through the local correction pathway. STAB3 was therefore not a new mechanism-search phase. Its goals were:

1. test whether the stabilization idea survived systematic sequence, depth, width, head-count, and precision scaling;
2. identify optimization and stochasticity boundaries without hiding negative cases;
3. freeze the training schedule rather than continually retune it;
4. compare stabilized QORLA against a controlled attention baseline under matched architecture and parameter count;
5. finish with a predeclared multi-split confirmation and systems Pareto audit.

The study deliberately separates three notions that are often conflated: numerical validity, optimization stability, and model quality. A run could have zero NaN/Inf events yet still collapse in accuracy. Likewise, smaller gradient norms were not treated as sufficient evidence of a better optimizer trajectory, because earlier experiments showed that large gradients can coexist with strong performance and small gradients with poor performance.

1.1 Frozen operator

For scalar query-like a_i , scalar key-like b_j , and value vector v_j ,

$$z_{ij} = b_j - a_i, \tag{4}$$

$$W_{ij} = e^{-\lambda|z_{ij}|}, \tag{5}$$

$$S_{0,i} = \sum_j W_{ij}, \tag{6}$$

$$\mu_i = \frac{\sum_j W_{ij} z_{ij}}{S_{0,i} + \epsilon}, \tag{7}$$

$$Q_i = \sum_j W_{ij} (z_{ij} - \mu_i)^2, \tag{8}$$

$$C_i = \sum_j W_{ij} (z_{ij} - \mu_i) v_j, \tag{9}$$

$$B_i = \frac{\sum_j W_{ij} v_j}{S_{0,i} + \epsilon}, \tag{10}$$

$$R_i = -\frac{\mu_i C_i}{Q_i + \rho + \epsilon}, \tag{11}$$

$$o_i = B_i + R_i. \tag{12}$$

The fixed constants used throughout the main controlled experiments were $\lambda = 4$, $\rho = 10^{-3}$, and $\epsilon = 10^{-8}$.

1.2 Training-time correction-gradient scaling

The causal intervention is

$$R_\alpha = \text{sg}(R) + \alpha(R - \text{sg}(R)). \quad (13)$$

Numerically, $R_\alpha = R$ for every α , so the forward pass is invariant. The backward pass becomes

$$\nabla_\theta R_\alpha = \alpha \nabla_\theta R, \quad (14)$$

and therefore

$$\boxed{\nabla_\theta o = \nabla_\theta B + \alpha(f) \nabla_\theta R.} \quad (15)$$

1.3 STAB3 experiment map

Stage	Main variable	Stabilizer state	Question
03-01	Sequence length	Permanent RSTOPALL	Does the correction stop-gradient remain numerically and computationally viable as N grows?
03-02	Depth	RSTOPALL $\rightarrow$ SG30 $\rightarrow$ SG10	Is permanent detachment too restrictive, and can a transient schedule retain stability without losing capacity?
03-03	Width	Frozen SG10	Does the schedule survive wider representations, including the first strong trajectory-sensitive regime?
03-04	Head count	Frozen SG10	Does stabilization transfer to realistic multi-head QORLA?
03-05	Precision	Frozen SG10	Does the method remain stable in BF16/FP16 training?
03-06	Optimization stress	Frozen SG10	Where are the LR, batch-size, and initialization boundaries?
03-07	Controlled attention comparison	Frozen SG10	Does stabilized QORLA outperform its own distance-softmax base branch?
03-08	Multi-split blind confirmation	Frozen SG10	Do the controlled conclusions survive fresh seeds and multiple data splits?

Table 1: STAB3 structure.

2 STAB3-01: Sequence-Length Scaling

2.1 03-01A: $N = 64, 128, 256, 512$

The first sequence-scaling study used a synthetic variable-length Gaussian-kernel teacher, width $D = 32$, input/output width 8, depth $L = 4$, Adam with learning rate 3×10^{-3} , and permanent RSTOPALL inherited from STAB2. Three seeds were run for each sequence length with 120 training steps.

The forward audit remained exact at every length and no invalid steps were observed. Validation-loss ratios were close to one, with no worsening trend as N increased. The principal systems effect was a consistent reduction in measured runtime and peak allocated GPU memory for RSTOPALL.

N	Validation ratio	Runtime ratio	Peak-memory ratio
64	0.929	0.824	0.938
128	1.009	0.698	0.899
256	1.024	0.648	0.881
512	1.007	0.627	0.869

Table 2: STAB3-01A, RSTOPALL/BASE ratios.

2.2 03-01B: $N = 1024, 2048$

The extension used fresh seeds 123–124 and 80 steps. At $N = 1024$, the mean validation-loss ratio was 0.9404, the runtime ratio 0.6071, and the peak-memory ratio 0.8530. At $N = 2048$, these became 0.9019, 0.6325, and 0.8510. Both seeds favored RSTOPALL in validation loss at both lengths.

N	Val. ratio	P95 grad ratio	Max grad ratio	Runtime ratio	Memory ratio
1024	0.9404	0.9252	0.8667	0.6071	0.8530
2048	0.9019	0.9168	0.8677	0.6325	0.8510

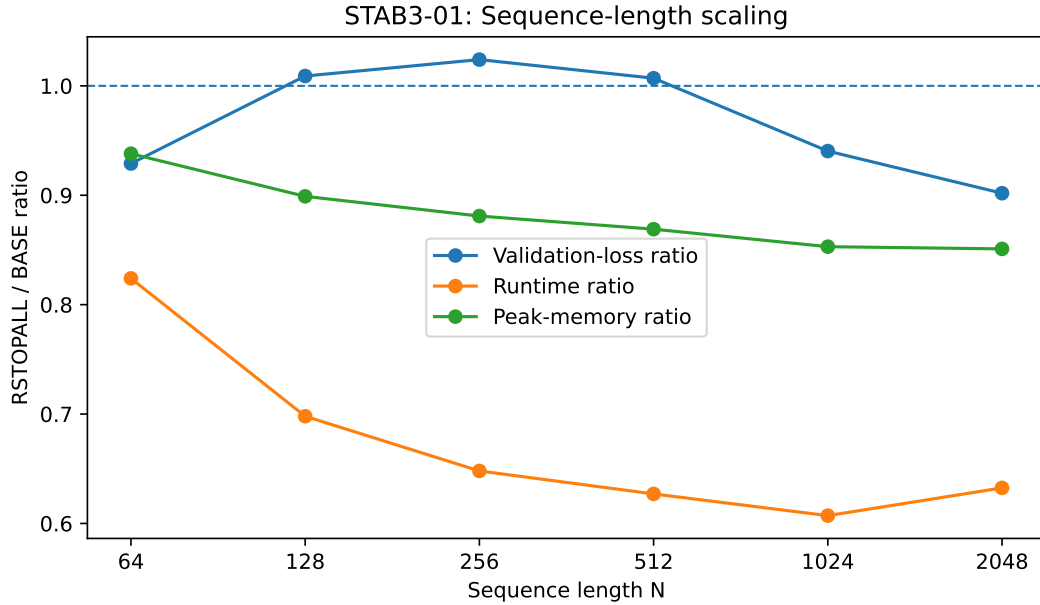
Table 3: STAB3-01B extension.

Figure 1: Sequence-length scaling. The ratios are empirical, implementation-specific measurements, not asymptotic complexity claims.

Verdict. Sequence scaling passed under the permanent stop-gradient intervention through $N = 2048$. The empirical 1024–2048 time exponents were approximately 0.936 for BASE and 0.997 for RSTOPALL, but batch size changed from 2 to 1 over that comparison. Consequently these measurements must not be presented as evidence for $O(N)$ complexity.

3 STAB3-02: Depth Scaling and Derivation of SG10

3.1 03-02A: Permanent RSTOPALL depth sweep

The Digits task used split seed 27182, width $D = 32$, sequence length $N = 64$, Adam at 3×10^{-3} , 12 epochs, and depths $L \in \{1, 2, 4, 6, 8, 12\}$. Seeds 125–127 were paired between BASE and permanent RSTOPALL.

Depth	BASE acc. (%)	RSTOPALL acc. (%)	Delta (pp)
1	93.52	70.65	-22.87
2	90.74	86.85	-3.89
4	86.85	74.35	-12.50
6	78.06	68.52	-9.54
8	77.13	64.91	-12.22
12	56.11 ± 31.03	59.72 ± 2.68	+3.61

Table 4: STAB3-02A: permanent detachment hurts otherwise stable depths.

Permanent RSTOPALL sharply reduced variability at $L = 12$ but imposed a strong capacity/optimization penalty at shallower and moderate depths. This ruled out permanent detachment as the final training rule.

3.2 03-02B: Undertraining audit

To test whether the 12-epoch comparison merely undertrained the detached model, $L = 4, 8, 12$ were trained for 30 epochs with the same seeds.

Depth	BASE loss	BASE acc.	RSTOPALL loss	RSTOPALL acc.
4	0.2273	93.43 ± 1.05	0.4348	86.67 ± 3.28
8	0.2036	93.98 ± 1.37	0.4453	87.31 ± 1.12
12	0.4997	82.50 ± 8.20	0.5336	83.06 ± 2.22

Table 5: STAB3-02B: longer training does not remove the permanent-detachment deficit at $L = 4, 8$.

At $L = 12$, RSTOPALL was strongly better early in training (44.63% vs 65.93% at epoch 12), but the BASE model caught up by epoch 30. This exposed the core tradeoff: early correction-gradient suppression is useful, permanent removal is not.

3.3 03-02C: SG30 warm-up

A forward-exact gradient restoration schedule was introduced: hold $\alpha = 0$ for the first 30%, linearly restore it during 30–60%, then use $\alpha = 1$. Fresh seeds 130–134 were used.

Depth	BASE	Permanent RSTOPALL	SG30
4	92.50	80.44	90.11
8	89.50	84.22	88.00
12	77.11	69.22	70.94

Table 6: STAB3-02C final accuracy (%). SG30 recovered much of the permanent-detachment deficit, but the protection window was too long.

3.4 03-02D: Final schedule selection and freeze

The final schedule experiment compared BASE, SG30, and a shorter SG10 schedule using fresh seeds 135–139 and depths 4, 8, and 12. No further schedule tuning was allowed after this experiment.

Depth	BASE final	SG10 final	SG30 final	BASE curve	SG10 curve
4	93.333 $\pm$ 1.416	92.333 $\pm$ 2.386	91.833 $\pm$ 4.081	80.009	77.680
8	89.333 $\pm$ 3.383	91.778 $\pm$ 2.130	90.611 $\pm$ 2.988	59.446	73.733
12	88.611 $\pm$ 4.714	90.500 $\pm$ 3.433	89.333 $\pm$ 2.535	56.500	71.139

Table 7: STAB3-02D. Curve accuracy is the mean accuracy across training epochs.

At epoch 12, the SG10 advantage was especially large at depth: 57.22 $\rightarrow$ 81.17% for $L = 8$ and 51.94 $\rightarrow$ 81.83% for $L = 12$. The P95 gradient ratios SG10/BASE were 0.829, 0.843, and 0.973 for $L = 4, 8, 12$, while the max-gradient ratios were 1.003, 0.512, and 0.838. SG30 produced a large $L = 4$ max-gradient ratio of 3.904, consistent with rebound after a longer protection interval.

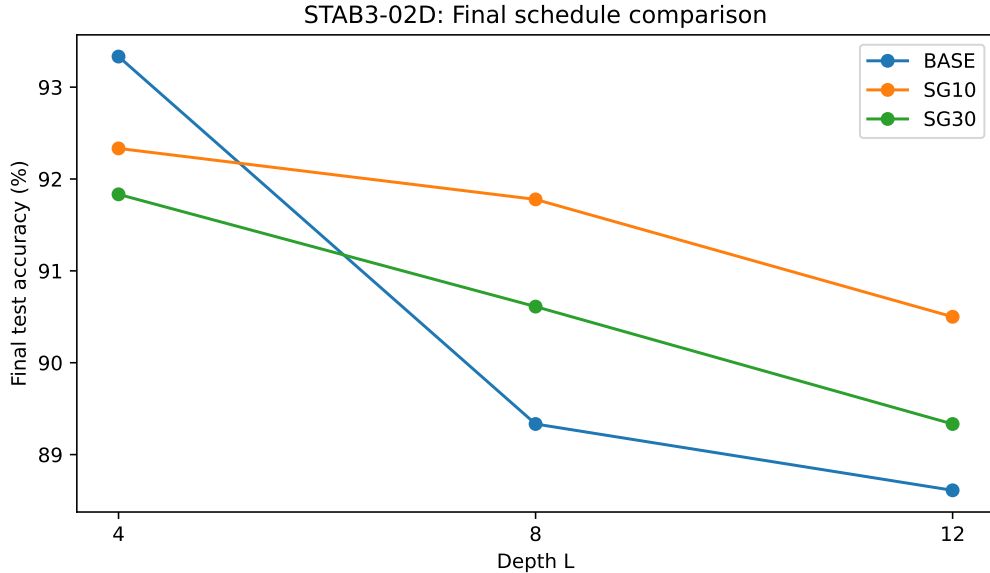


Figure 2: STAB3-02D final schedule comparison.

Frozen rule. SG10 was selected because it preserved the early stabilizing effect while restoring ordinary QORLA gradients quickly enough to avoid the capacity penalty of permanent detachment. At $L = 4$ it was slightly worse than BASE, so the claim is explicitly depth-dependent rather than universal.

4 STAB3-03: Width Scaling

4.1 03-03A: $D = 16, 32, 64, 128$

The architecture used depth $L = 8$, sequence length 64, split seed 27182, Adam at 3×10^{-3} , 30 epochs, and seeds 140–142. Parameter count increased from 21,242 at $D = 16$ to 1,201,946 at $D = 128$.

D	BASE final	SG10 final	Delta	BASE curve	SG10 curve
16	90.278 ± 2.546	93.426 ± 1.052	+3.148	67.688	74.003
32	90.093 ± 1.530	92.685 ± 1.628	+2.593	68.870	70.719
64	57.407 ± 41.090	78.426 ± 11.077	+21.019	40.241	54.546
128	59.630 ± 11.760	51.481 ± 30.962	-8.148	35.216	42.639

Table 8: STAB3-03A width scaling.

At $D = 64$, SG10 substantially reduced the instability visible in BASE. At $D = 128$, however, one severe SG10 trajectory collapse reversed the final mean despite a better mean training curve. This triggered one predeclared diagnostic rather than another schedule search.

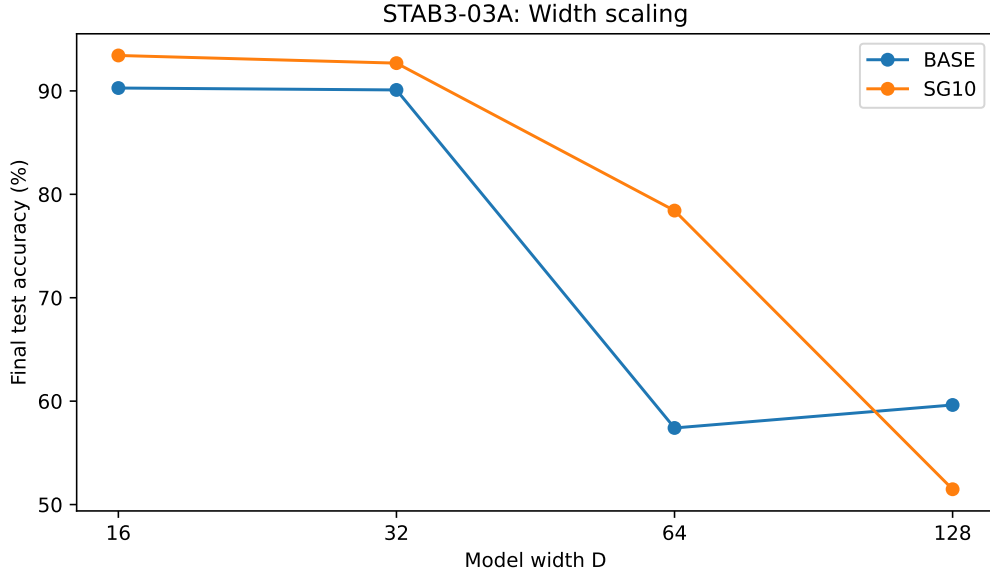


Figure 3: Width scaling. The $D = 128$ point exposes trajectory sensitivity rather than a monotonic width law.

4.2 03-03B: $D = 128$ width-by-learning-rate diagnostic

Fresh seeds 143–147 were run at $\eta \in \{3 \times 10^{-3}, 10^{-3}\}$ with the same $D = 128, L = 8$ architecture.

LR	Method	Final acc.	Std.	Min	Curve acc.	Late drop
0.001	BASE	72.833	25.900	27.500	49.909	6.000
0.001	SG10	77.111	17.475	56.389	55.189	2.167
0.003	BASE	51.278	15.800	31.667	27.439	1.667
0.003	SG10	65.556	17.223	51.389	45.333	2.222

Table 9: STAB3-03B summary.

At 10^{-3} SG10 improved mean accuracy by +4.278 pp with 3/5 wins and reduced mean best-to-final deterioration by 3.833 pp. At 3×10^{-3} it improved mean accuracy by +14.278 pp with 4/5 wins. Nevertheless severe negative per-seed deltas remained, including approximately -29.72 pp at 10^{-3} and -19.44 pp at 3×10^{-3} . Thus the $D = 128$ issue was partly a width-by-learning-rate interaction, but not completely explained by learning rate.

Verdict. Width scaling was closed as mixed but positive. SG10 transfers robustly through $D = 64$ and improves $D = 128$ on average in fresh validation, yet very wide QORLA remains trajectory-sensitive. No further width or schedule tuning was permitted.

5 STAB3-04: Multi-Head Scaling

The multi-head study fixed total model width $D = 64$, depth $L = 8$, sequence length 64, Adam at 3×10^{-3} , 30 epochs, and seeds 148–150. The number of QORLA heads was $H \in \{1, 2, 4, 8\}$, so head dimension decreased from 64 to 8. Parameter count rose only slightly from 339,354 to 346,634, so this is best described as fixed total width with approximately matched parameter count.

H	d_h	BASE acc.	SG10 acc.	Delta	Wins	BASE std.	SG10 std.
1	64	45.926	66.111	+20.185	1/3	31.655	19.151
2	32	67.037	78.611	+11.574	3/3	11.595	7.185
4	16	84.537	87.870	+3.333	2/3	4.009	1.123
8	8	60.185	91.667	+31.481	3/3	22.787	3.413

Table 10: STAB3-04A multi-head scaling.

At $H = 8$, all three paired runs improved and even the worst paired delta was +15.278 pp. The P95 gradient ratio was 0.425 and the max-gradient ratio 0.215. Peak memory was essentially identical between BASE and SG10, while runtime differed by only a few percent.

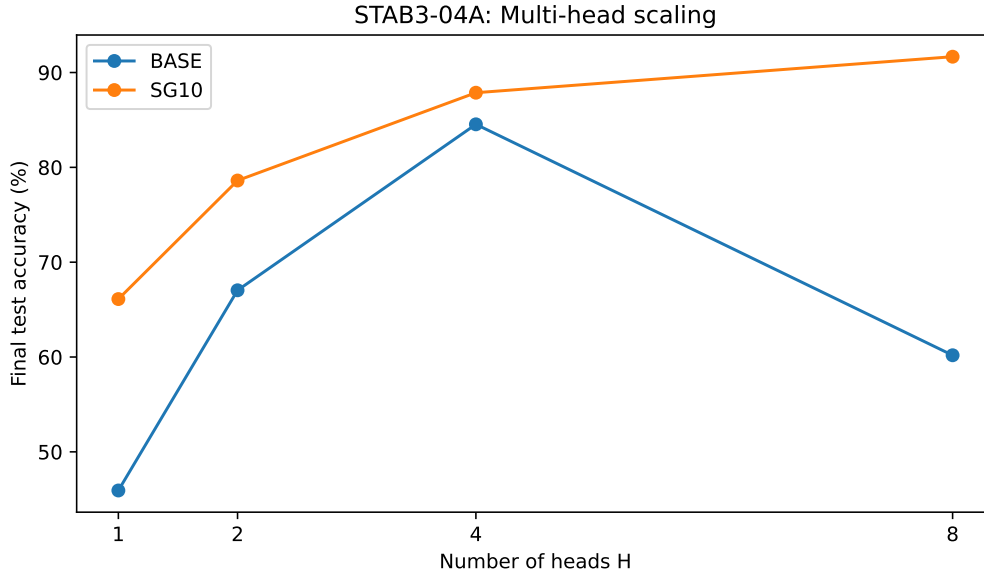


Figure 4: Multi-head scaling. The strongest gain appears at $H = 8$.

Verdict. Multi-head scaling strongly passed. The schedule was not specific to the single-head geometry used in early experiments.

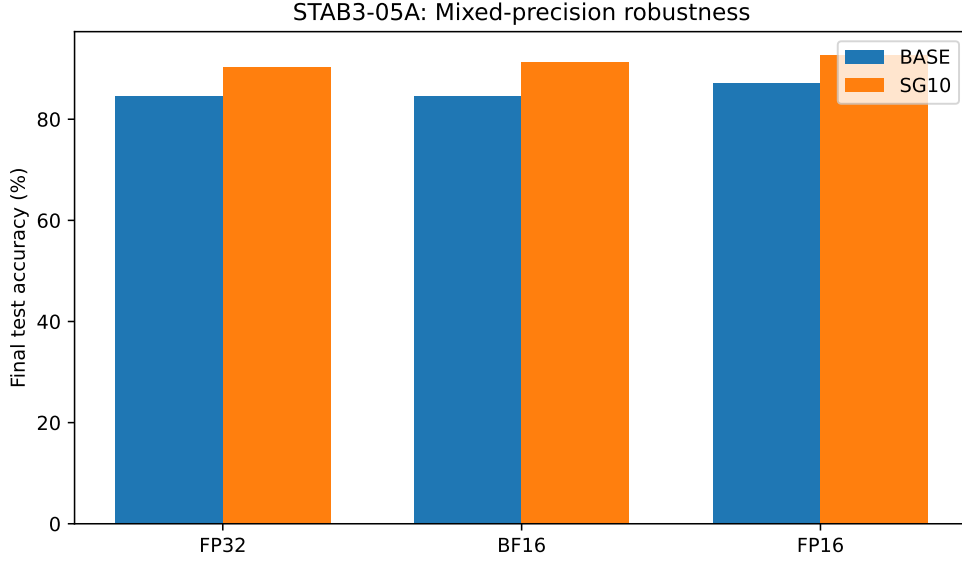
6 STAB3-05: Mixed-Precision Stability

The frozen $D = 64$, $L = 8$, $H = 8$ model was trained in FP32, BF16, and FP16 on a Tesla T4 using seeds 151–153. The same split, batch size 64, learning rate 3×10^{-3} , and 30-epoch budget were used.

Precision	BASE acc.	SG10 acc.	Delta	SG10 wins	BASE invalid/overflow	SG10 invalid/overflow
FP32	84.444	90.278	+5.833	2/3 (one tie)	0/0	0/0
BF16	84.537	91.204	+6.667	3/3	0/0	0/0
FP16	87.130	92.685	+5.556	3/3	18/18	9/9

Table 11: STAB3-05A mixed-precision results. FP16 invalid-gradient events coincide with dynamic-loss-scaling overflow events and were recoverable rather than persistent run failures.

BF16 provided the cleanest reduced-precision result: zero invalid steps, zero overflows, and a +6.667 pp mean gain. FP16 also trained successfully, but transient overflow events were observed and handled by GradScaler. SG10 halved the observed FP16 overflow count from 18 to 9 across the three seeds.

**Figure 5:** Mixed-precision final accuracy.

The Tesla T4 did not show a BF16 throughput advantage, so this section establishes numerical/training robustness rather than a mixed-precision systems speedup.

7 STAB3-06: Optimization Stress

7.1 03-06A: Learning-rate stress

The BF16 $D = 64, L = 8, H = 8$ configuration was run with fresh seeds 154–156 at learning rates 10^{-3} , 3×10^{-3} , 6×10^{-3} , and 10^{-2} . All runs remained finite.

LR	BASE acc.	SG10 acc.	Delta	SG10 wins	BASE curve	SG10 curve
0.001	76.204	90.648	+14.444	3/3	48.645	69.012
0.003	81.389	90.370	+8.981	3/3	44.528	71.056
0.006	82.222	50.000	-32.222	0/3	50.685	48.083
0.010	45.556	25.741	-19.815	0/3	23.991	27.728

Table 12: STAB3-06A learning-rate stress.

At 6×10^{-3} , SG10 was initially ahead but deteriorated after the correction gradient had been fully restored. The SG10 max-gradient ratio rose to about 5.05 relative to BASE. At 10^{-2} it rose to about 11.67, and one SG10 run crossed the predefined $< 20\%$ collapse threshold. This establishes a clear boundary: SG10 improves the normal operating region but does not expand the maximum stable-learning-rate envelope.

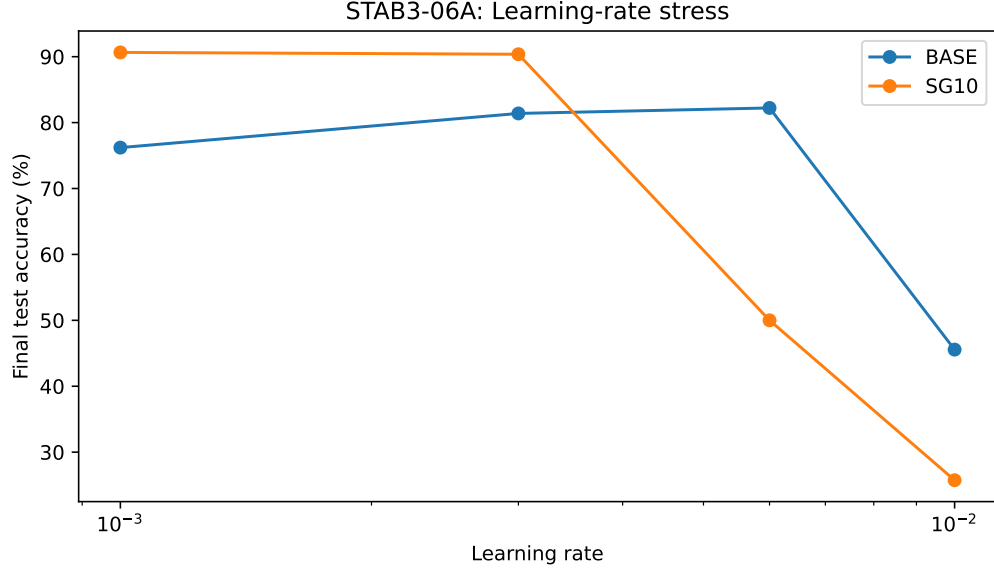


Figure 6: Learning-rate stress. The reversal above the normal operating range is retained as a failure boundary.

7.2 03-06B: Batch-size by Q/K initialization stress

The learning rate was fixed back at 3×10^{-3} . Batch sizes 16, 64, 256 and targeted Q/K initialization scales $\times 1$ and $\times 2$ were tested with seeds 157–159.

Batch	Q/K scale	BASE acc.	SG10 acc.	Delta	SG10 wins
16	1	82.870	55.000	-27.870	0/3
16	2	82.407	64.907	-17.500	0/3
64	1	85.093	92.685	+7.593	3/3
64	2	82.500	90.093	+7.593	3/3
256	1	49.167	78.241	+29.074	3/3
256	2	62.130	87.685	+25.556	3/3

Table 13: STAB3-06B final accuracy.

At batch size 16, SG10 learned much faster early but deteriorated late. For Q/K scale 1, its mean accuracy was already 82.41% at epoch 12 versus 38.52% for BASE, yet final accuracy fell to 55.00%. The same pattern appeared for Q/K scale 2. In contrast, both initialization scales remained strongly positive at batch sizes 64 and 256.

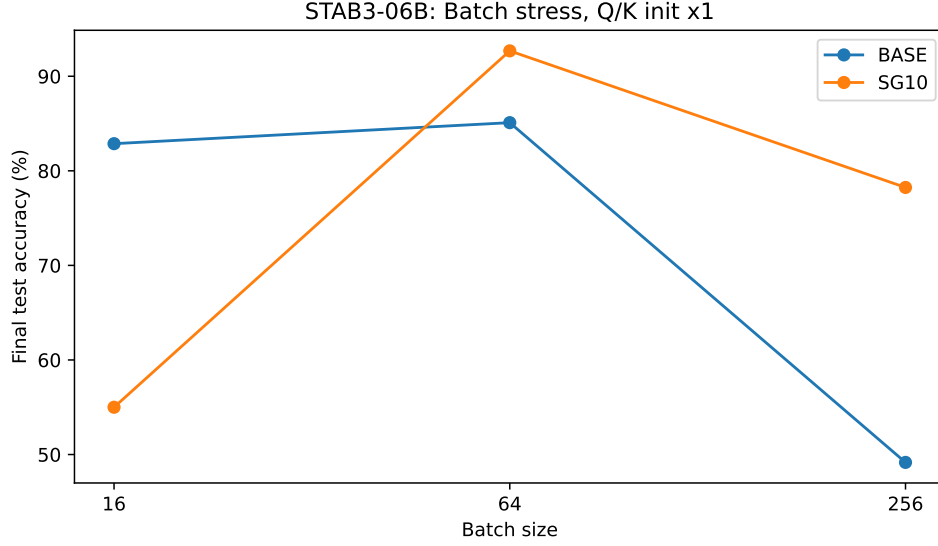


Figure 7: Batch-size stress with normal Q/K initialization.

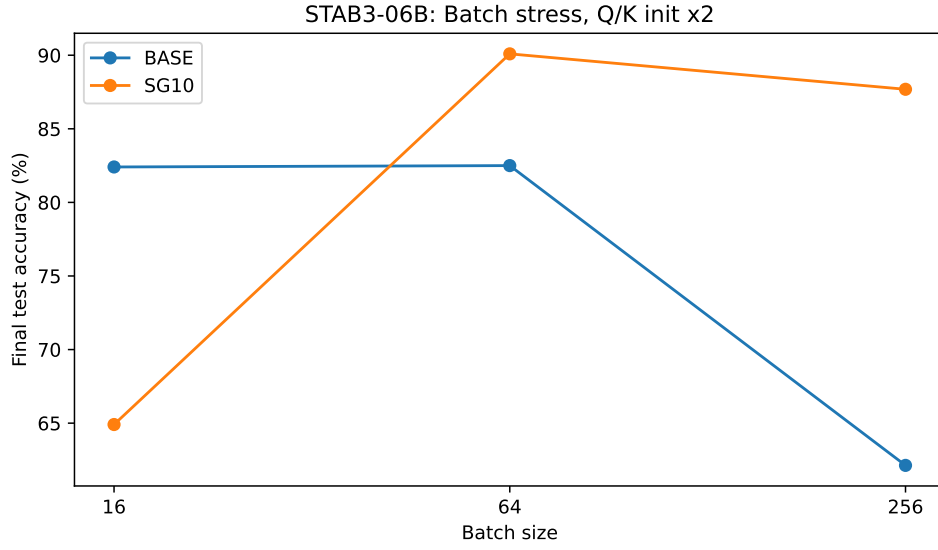


Figure 8: Batch-size stress with doubled Q/K initialization.

Verdict. The main stochasticity boundary is very small batch size, not the tested Q/K initialization scale. The phase was closed without modifying SG10 to rescue batch size 16.

8 STAB3-07: Controlled Comparative Attention Stability

A controlled comparator was constructed from the QORLA base branch. For each head,

$$A_{ijh} = \text{softmax}_j(-\lambda|k_{jh} - q_{ih}|), \quad B_{ih} = \sum_j A_{ijh} v_{jh}. \quad (16)$$

The distance-softmax output matched the QORLA B branch to numerical precision (maximum gap 2.38×10^{-7} in STAB3-07A). All three methods used exactly 346,634 parameters and identical initialization, minibatch ordering, Q/K/V projections, output projection, and FFN structure.

Fresh seeds 160–164 were used.

Method	Final acc.	Std.	Curve acc.	P95 grad	Runtime (s)	Peak MB
DIST-SOFTMAX	89.167	2.088	61.424	3.016	41.949	546.833
QORLA-BASE	85.222	4.192	42.522	18.127	75.709	588.317
QORLA-SG10	92.722	2.082	71.170	5.859	73.688	589.317

Table 14: STAB3-07A controlled comparison.

Pairwise, SG10 beat BASE by +7.500 pp with 5/5 wins and improved mean training-curve accuracy by +28.648 pp. Against DIST-SOFTMAX, SG10 gained +3.556 pp with 5/5 wins and +9.746 pp in curve accuracy. In contrast, QORLA-BASE was 3.944 pp worse than DIST-SOFTMAX and won only 1/5 paired runs.

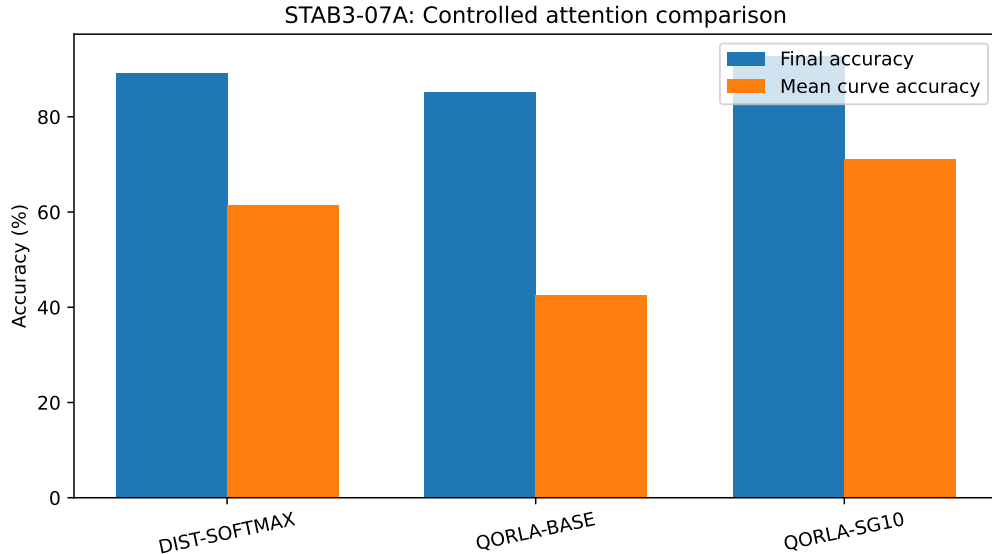


Figure 9: Controlled attention comparison. The correction branch is harmful when trained naively but beneficial under SG10.

The key mechanistic interpretation is therefore not that adding R is automatically useful. Rather, R appears to contain useful representational capacity whose unrestricted early backward dynamics can prevent that capacity from being realized.

The systems limitation is equally important. SG10 required about $1.76\times$ the runtime and $1.08\times$ the peak memory of the distance-softmax control in this implementation. STAB3 therefore does not support a speed or memory superiority claim over softmax-like aggregation.

9 STAB3-08: Final Multi-Split Confirmation and Pareto Audit

The final experiment froze all decisions: $D = 64$, $L = 8$, $H = 8$, batch size 64, Adam at 3×10^{-3} , BF16, 30 epochs, and the same SG10 schedule. Three data splits (2027, 31415, 92653) and three fresh model seeds (165–167) produced nine paired comparisons per method and 27 total runs. Method execution order was shuffled within each split/seed cell.

The parameter count was 346,634. QORLA forward exactness remained 0.0 and the B /distance-softmax gap was 4.77×10^{-7} . No method produced an invalid step or a predefined collapse.

9.1 Overall final results

Method	Final acc.	Std.	Min	Max	Curve acc.	Runtime	Throughput	Peak MB
DIST-SOFTMAX	89.969	1.709	86.667	92.222	64.514	41.816	1030.993	501.770
QORLA-BASE	81.235	11.719	50.556	87.500	49.762	75.737	569.208	588.254
QORLA-SG10	91.574	1.667	88.333	93.611	72.850	73.621	585.569	589.254

Table 15: STAB3-08 final blind summary.

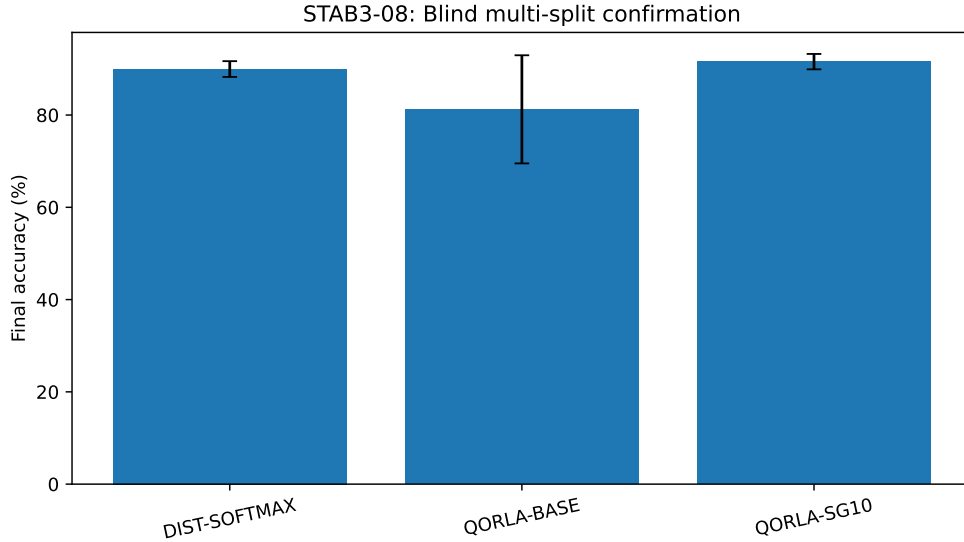


Figure 10: Final multi-split accuracy with standard deviation across nine runs.

9.2 Paired comparisons

Comparison	Mean delta	95% bootstrap CI	Wins	Worst delta	Curve delta	Time ratio	Throughput ratio	Memory ratio
SG10 / BASE	+10.340	[4.907, 18.858]	9/9	+1.667	+23.087	0.972	1.029	1.002
SG10 / DIST-SOFTMAX	+1.605	[0.309, 2.994]	7/9	-1.111	+8.335	1.761	0.568	1.174
BASE / DIST-SOFTMAX	-8.735	[-17.531, -3.241]	0/9	-41.667	-14.752	1.811	0.552	1.172

Table 16: Final paired comparisons. Confidence intervals are nonparametric bootstrap intervals over the nine paired accuracy differences.

9.3 Split-level behavior

Split	DIST-SOFTMAX	QORLA-BASE	QORLA-SG10
2027	90.370	85.370	90.833
31415	91.019	74.074	91.111
92653	88.519	84.259	92.778

Table 17: Mean final accuracy by data split.

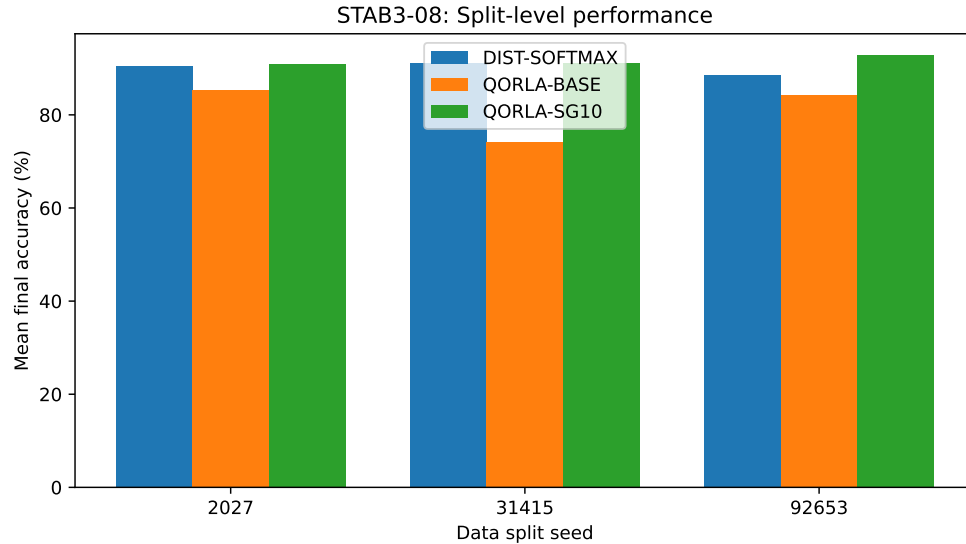


Figure 11: Split-level performance. SG10 has the highest split mean in all three splits.

The split IDs had appeared earlier in the broader project, so this experiment should be described as a predeclared multi-split confirmation rather than a globally untouched benchmark. The fresh model seeds and fixed STAB3 protocol nevertheless provide a strong final within-project validation.

9.4 Systems Pareto

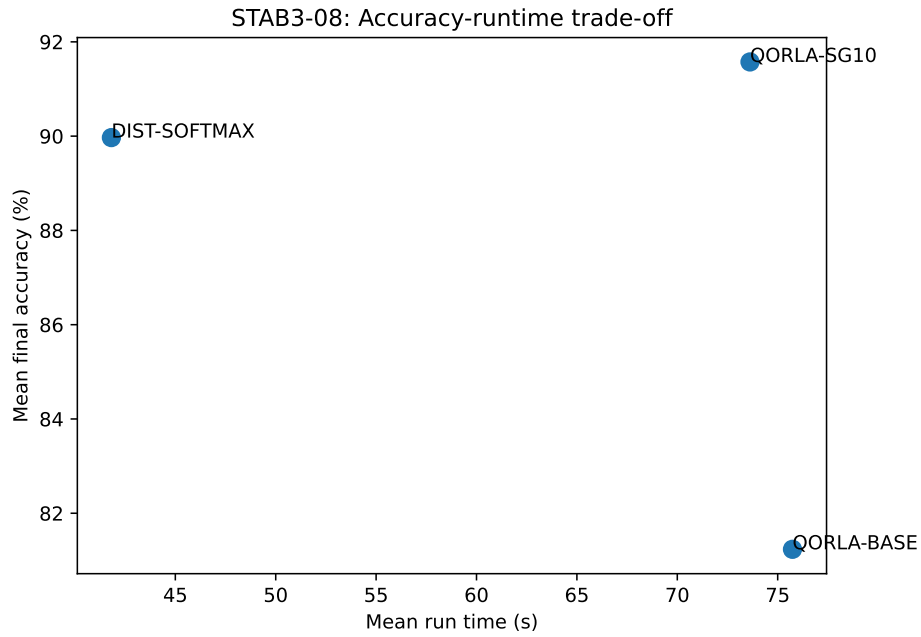


Figure 12: Accuracy-runtime trade-off. DIST-SOFTMAX is faster; SG10 is more accurate.

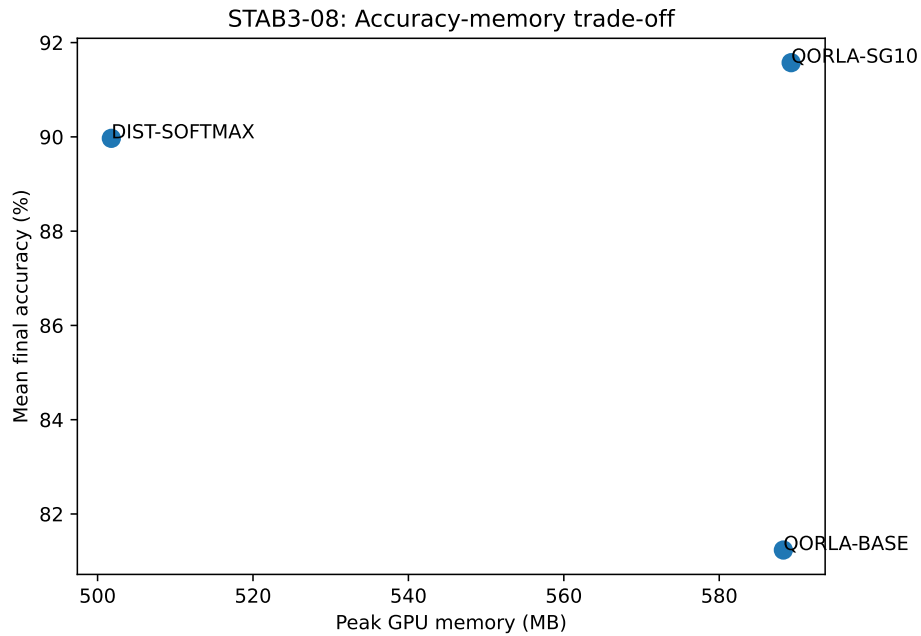


Figure 13: Accuracy-memory trade-off. The controlled QORLA implementation uses more memory than DIST-SOFTMAX.

Final STAB3 verdict. SG10 decisively stabilizes QORLA relative to ordinary training and produces a modest but repeatable accuracy advantage over the controlled distance-softmax base branch. That accuracy advantage comes at increased operator cost in the present implementation.

10 Cross-Experiment Synthesis

10.1 What transferred successfully

- **Depth:** the transient schedule preserved early stabilization without the permanent capacity loss of RSTOPALL; the strongest benefits appeared at $L = 8$ and $L = 12$.
- **Width:** benefits were robust through $D = 64$ and positive on average at $D = 128$, although wide trajectories remained seed-sensitive.
- **Multi-head structure:** the method transferred strongly to $H = 2, 4, 8$, with the largest average gain at $H = 8$.
- **Reduced precision:** BF16 remained fully clean and FP16 remained trainable under dynamic loss scaling.
- **Standard and large batches:** batch sizes 64 and 256 produced consistent gains under both normal and doubled Q/K initialization scales.
- **Controlled comparator:** SG10 beat the distance-softmax base branch in both the five-seed controlled comparison and the final nine-pair multi-split confirmation.

10.2 Observed failure boundaries

Boundary	Observation	Interpretation
Permanent detachment	Strong accuracy loss at stable shallow/moderate depth	Correction gradients are useful after early training; permanent removal is overregularizing.
Very wide $D = 128$	High seed sensitivity and severe paired outliers	Width amplifies trajectory dependence; LR interaction is important but not sufficient to explain all failures.
Aggressive $\text{LR} \geq 6 \times 10^{-3}$	SG10 reverses from beneficial to harmful	Early protection does not expand the maximum stable optimizer-step envelope.
Batch size 16	Strong early learning followed by late deterioration	High stochastic gradient noise interacts poorly with restoration of the full correction gradient.
FP16	Recoverable overflow events	Numerical range is tighter; dynamic loss scaling is required.
Systems cost	SG10 slower and more memory-intensive than DIST-SOFTMAX	Current evidence is about stability/quality, not implementation efficiency.

Table 18: STAB3 failure envelope.

10.3 What the gradient evidence does and does not show

Gradient magnitudes often fell under SG10, especially in successful deep and multi-head settings, but gradient norm was not a universal predictor of quality. At $D = 64$ in the width study, SG10 had larger P95 and max gradients than BASE while performing much better. At aggressive learning rates, SG10 eventually produced much larger gradients and deteriorated. These results support a trajectory- and pathway-specific interpretation rather than a simple “smaller gradient is better” rule.

11 Final Frozen QORLA Training Specification

The version carried into STAB4 is frozen as follows.

11.1 Forward operator

$$\begin{aligned}
 z_{ij} &= b_j - a_i, \\
 W_{ij} &= e^{-\lambda|z_{ij}|}, \\
 S_{0,i} &= \sum_j W_{ij}, \\
 \mu_i &= \frac{\sum_j W_{ij} z_{ij}}{S_{0,i} + \epsilon}, \\
 Q_i &= \sum_j W_{ij} (z_{ij} - \mu_i)^2, \\
 C_i &= \sum_j W_{ij} (z_{ij} - \mu_i) v_j, \\
 B_i &= \frac{\sum_j W_{ij} v_j}{S_{0,i} + \epsilon}, \\
 R_i &= -\frac{\mu_i C_i}{Q_i + \rho + \epsilon}, \\
 o_i &= B_i + R_i.
 \end{aligned} \tag{17}$$

11.2 Training rule

$$o_i^{\text{train}} = B_i + \text{sg}(R_i) + \alpha(f) [R_i - \text{sg}(R_i)] \tag{18}$$

with

$$\alpha(f) = \begin{cases} 0, & f \leq 0.10, \\ (f - 0.10)/0.20, & 0.10 < f \leq 0.30, \\ 1, & f > 0.30. \end{cases} \tag{19}$$

At inference, no stop-gradient schedule is required; the operator is simply $o = B + R$.

12 Claims Supported by STAB3

The following statements are supported by the controlled evidence in this report:

1. QORLA’s correction branch can generate severe optimization instability when its gradient is unrestricted from the start of training.
2. Permanent correction-gradient removal is too aggressive in stable regimes and can reduce useful modeling capacity.
3. A forward-exact transient correction-gradient schedule can substantially improve training trajectories and final accuracy in deep, multi-head, reduced-precision, and several width/batch regimes.

4. The final SG10 rule can outperform a parameter-matched distance-softmax control in controlled Digits experiments.
5. The stabilized model retains explicit failure boundaries, especially at aggressive learning rates and very small batches.

13 Claims Not Supported Yet

STAB3 does **not** establish the following:

- that QORLA is universally better than standard Transformer dot-product softmax;
- that QORLA is faster or more memory-efficient than optimized softmax or FlashAttention;
- that the operator has linear complexity in sequence length;
- that SG10 improves every shallow or low-noise regime;
- that the method generalizes to real language datasets, pretrained LLMs, vision transformers, or multimodal models;
- that the observed bootstrap intervals constitute a definitive external statistical benchmark; they summarize the nine paired controlled runs only.

14 Transition to STAB4

STAB3 is complete. The next phase must leave the controlled Digits/synthetic environment and test the frozen QORLA formulation on real sequence data and real Transformer architectures. The planned transition is:

1. real text classification with QORLA inserted into a Transformer encoder;
2. language modeling with perplexity, tokens/sec, memory, and long-context evaluation;
3. matched comparison with standard Transformer softmax and selected established efficient-attention methods;
4. long-context real-text experiments;
5. pretrained small language-model integration and, if compute permits, extension toward larger LLMs;
6. later multimodal transfer.

The schedule, constants, and stabilization rule are frozen before those external evaluations. STAB4 should therefore evaluate transfer, not continue schedule search.

A Condensed Run-Level Evidence

This appendix preserves the most important paired run-level outcomes from the later STAB3 experiments. Values are final test accuracy in percent unless stated otherwise.

A.1 STAB3-03B: $D = 128$ diagnostic

Seed	LR	Method	Final	Best	Best epoch	Late drop
143	0.003	BASE	31.667	33.056	27	1.389
143	0.003	SG10	51.389	51.389	21	0.000
143	0.001	BASE	86.667	89.167	26	2.500
143	0.001	SG10	88.056	90.833	29	2.778
144	0.003	BASE	75.556	76.944	28	1.389
144	0.003	SG10	56.111	61.111	23	5.000
144	0.001	BASE	85.278	87.500	29	2.222
144	0.001	SG10	56.389	56.389	30	0.000
145	0.003	BASE	53.056	53.056	30	0.000
145	0.003	SG10	81.111	84.722	28	3.611
145	0.001	BASE	89.444	91.389	29	1.944
145	0.001	SG10	59.722	66.667	27	6.944
146	0.003	BASE	48.611	48.611	29	0.000
146	0.003	SG10	51.944	54.444	29	2.500
146	0.001	BASE	27.500	50.000	29	22.500
146	0.001	SG10	91.389	91.944	29	0.556
147	0.003	BASE	47.500	53.056	29	5.556
147	0.003	SG10	87.222	87.222	30	0.000
147	0.001	BASE	75.278	76.111	29	0.833
147	0.001	SG10	90.000	90.556	25	0.556

A.2 STAB3-04A: Multi-head runs

Heads	Seed	Method	Final	Best	Late drop
1	148	BASE	10.0	10.8	0.8
1	148	SG10	86.4	86.9	0.6
1	149	BASE	69.7	75.8	6.1
1	149	SG10	63.6	72.8	9.2
1	150	BASE	58.1	60.8	2.8
1	150	SG10	48.3	48.3	0.0
2	148	BASE	76.7	80.6	3.9
2	148	SG10	86.4	86.7	0.3
2	149	BASE	70.3	74.7	4.4
2	149	SG10	77.2	79.2	1.9
2	150	BASE	54.2	61.7	7.5
2	150	SG10	72.2	73.6	1.4

Heads	Seed	Method	Final	Best	Late drop
4	148	BASE	89.2	89.4	0.3
4	148	SG10	87.2	89.4	2.2
4	149	BASE	82.2	82.8	0.6
4	149	SG10	87.2	89.2	1.9
4	150	BASE	82.2	83.6	1.4
4	150	SG10	89.2	89.4	0.3
8	148	BASE	68.3	68.3	0.0
8	148	SG10	87.8	90.6	2.8
8	149	BASE	77.8	79.2	1.4
8	149	SG10	93.1	94.2	1.1
8	150	BASE	34.4	40.6	6.1
8	150	SG10	94.2	94.4	0.3

A.3 STAB3-05A: Precision runs

Seed	Precision	Method	Final	Best	Overflow events
151	FP32	BASE	79.7	79.7	0
151	FP32	SG10	88.1	91.4	0
151	BF16	BASE	85.0	86.1	0
151	BF16	SG10	90.0	92.2	0
151	FP16	BASE	85.8	86.9	5
151	FP16	SG10	89.7	91.1	4
152	FP32	BASE	84.4	84.4	0
152	FP32	SG10	93.6	95.0	0
152	BF16	BASE	83.1	83.6	0
152	BF16	SG10	93.9	93.9	0
152	FP16	BASE	85.6	88.3	7
152	FP16	SG10	95.3	95.3	3
153	FP32	BASE	89.2	90.3	0
153	FP32	SG10	89.2	91.9	0
153	BF16	BASE	85.6	85.6	0
153	BF16	SG10	89.7	92.8	0
153	FP16	BASE	90.0	90.0	6
153	FP16	SG10	93.1	93.6	2

A.4 STAB3-06A: Learning-rate stress runs

Seed	LR	Method	Final	Best	Late drop
154	0.001	BASE	86.7	91.9	5.3
154	0.001	SG10	92.5	93.6	1.1
154	0.003	BASE	82.5	82.5	0.0
154	0.003	SG10	91.1	92.8	1.7

Seed	LR	Method	Final	Best	Late drop
154	0.006	BASE	80.0	85.8	5.8
154	0.006	SG10	76.4	92.2	15.8
154	0.010	BASE	60.6	63.3	2.8
154	0.010	SG10	28.1	50.0	21.9
155	0.001	BASE	80.0	80.6	0.6
155	0.001	SG10	89.4	90.8	1.4
155	0.003	BASE	75.8	81.1	5.3
155	0.003	SG10	91.4	92.2	0.8
155	0.006	BASE	82.2	85.3	3.1
155	0.006	SG10	20.6	50.8	30.3
155	0.010	BASE	43.9	46.9	3.1
155	0.010	SG10	30.0	61.7	31.7
156	0.001	BASE	61.9	62.5	0.6
156	0.001	SG10	90.0	92.5	2.5
156	0.003	BASE	85.8	89.4	3.6
156	0.003	SG10	88.6	92.2	3.6
156	0.006	BASE	84.4	84.4	0.0
156	0.006	SG10	53.1	69.7	16.7
156	0.010	BASE	32.2	41.4	9.2
156	0.010	SG10	19.2	69.7	50.6

A.5 STAB3-06B: Batch/init stress runs

Batch	Q/K	Seed	Method	Final	Best	Late drop
16	1	157	BASE	72.5	81.4	8.9
16	1	157	SG10	54.2	90.8	36.7
16	1	158	BASE	91.1	91.7	0.6
16	1	158	SG10	49.7	85.0	35.3
16	1	159	BASE	85.0	87.2	2.2
16	1	159	SG10	61.1	88.1	26.9
16	2	157	BASE	76.7	79.4	2.8
16	2	157	SG10	52.5	89.4	36.9
16	2	158	BASE	88.1	88.1	0.0
16	2	158	SG10	86.4	87.5	1.1
16	2	159	BASE	82.5	85.3	2.8
16	2	159	SG10	55.8	85.0	29.2
64	1	157	BASE	85.8	85.8	0.0
64	1	157	SG10	96.9	96.9	0.0
64	1	158	BASE	81.4	86.9	5.6
64	1	158	SG10	86.9	92.2	5.3
64	1	159	BASE	88.1	92.5	4.4
64	1	159	SG10	94.2	94.7	0.6
64	2	157	BASE	86.4	86.4	0.0
64	2	157	SG10	91.4	91.9	0.6

Batch	Q/K	Seed	Method	Final	Best	Late drop
64	2	158	BASE	76.7	83.3	6.7
64	2	158	SG10	89.2	89.7	0.6
64	2	159	BASE	84.4	87.2	2.8
64	2	159	SG10	89.7	92.5	2.8
256	1	157	BASE	30.0	32.8	2.8
256	1	157	SG10	85.3	88.6	3.3
256	1	158	BASE	47.5	47.5	0.0
256	1	158	SG10	65.8	65.8	0.0
256	1	159	BASE	70.0	70.0	0.0
256	1	159	SG10	83.6	83.9	0.3
256	2	157	BASE	67.5	67.5	0.0
256	2	157	SG10	91.1	91.1	0.0
256	2	158	BASE	52.2	52.2	0.0
256	2	158	SG10	86.4	87.5	1.1
256	2	159	BASE	66.7	66.7	0.0
256	2	159	SG10	85.6	86.1	0.6

A.6 STAB3-07A: Controlled comparison runs

Seed	Method	Final	Best	Late drop	Curve acc.
160	DIST-SOFTMAX	86.944	86.944	0.000	58.241
160	QORLA-BASE	90.000	90.000	0.000	48.694
160	QORLA-SG10	90.556	90.556	0.000	67.972
161	DIST-SOFTMAX	91.389	92.222	0.833	64.296
161	QORLA-BASE	79.167	79.167	0.000	33.880
161	QORLA-SG10	91.667	91.667	0.000	71.565
162	DIST-SOFTMAX	88.333	88.333	0.000	60.556
162	QORLA-BASE	87.222	87.222	0.000	41.093
162	QORLA-SG10	92.500	94.444	1.944	73.843
163	DIST-SOFTMAX	91.389	92.500	1.111	64.361
163	QORLA-BASE	86.667	86.667	0.000	44.000
163	QORLA-SG10	96.111	96.111	0.000	73.732
164	DIST-SOFTMAX	87.778	89.722	1.944	59.667
164	QORLA-BASE	83.056	83.056	0.000	44.944
164	QORLA-SG10	92.778	94.167	1.389	68.741

A.7 STAB3-08: All nine SG10 vs distance-softmax blind pairs

Split	Seed	SG10 acc.	DIST acc.	Delta	Curve delta
2027	165	89.722	89.167	+0.556	+5.046
2027	166	91.667	90.000	+1.667	+4.111
2027	167	91.111	91.944	-0.833	+9.583
31415	165	92.500	92.222	+0.278	+13.222
31415	166	88.333	89.444	-1.111	+9.907
31415	167	92.500	91.389	+1.111	+5.639
92653	165	93.056	89.444	+3.611	+13.667
92653	166	91.667	86.667	+5.000	+5.685
92653	167	93.611	89.444	+4.167	+8.157

Table 25: Final SG10 vs distance-softmax paired evidence.

B Reproducibility Notes

- Most Digits studies used stratified train/test splitting with split seed 27182; the final confirmation used split seeds 2027, 31415, and 92653.
- Paired methods within an experiment shared initialization and minibatch ordering unless explicitly stated otherwise.
- Later STAB3 experiments used a Tesla T4 and BF16 where indicated.
- Forward-equality audits were performed repeatedly when stop-gradient schedules were used.
- Invalid-step counts refer to NaN/Inf loss or gradient events as defined in the experimental cells. FP16 overflow events in STAB3-05 were recoverable dynamic-loss-scaling events.
- The final method is frozen at SG10; later work should not reinterpret STAB4 as another schedule-tuning phase.

C Final Record

STABILITY TEST 3 - CLOSED

Frozen forward operator: $o = B + R$.

Frozen training rule: SG10 correction-gradient schedule.

No additional STAB3 tuning after STAB3-08.

Next phase: real-world Transformer, language-model, and external-baseline evaluation.